Exp 7 Flow control at Data link layer.

Aim: Write a program to implement flow control at data link layer using Sliding window protocol. Simulate the flow of frame from one node to another

Create a Sender program with following features

1 Input Window Size from the user.

2. Input a test message from the user

3. Consider 1 character per frame

4. Create a frame with following fields

5 Send the frame

6. Wait for the acknowledgement from the Receiver

Create a receive file with following

1. Reader a file called Sender-Buff

2. Check the frame no.

3. If the Frame no. are as expected write the appropriate ACK no in the Receiver Buffer file

```
def Sliding_window (total_frames, window_size
        loss_frame = -1);
    if window_size < 1:
        window_size = 1

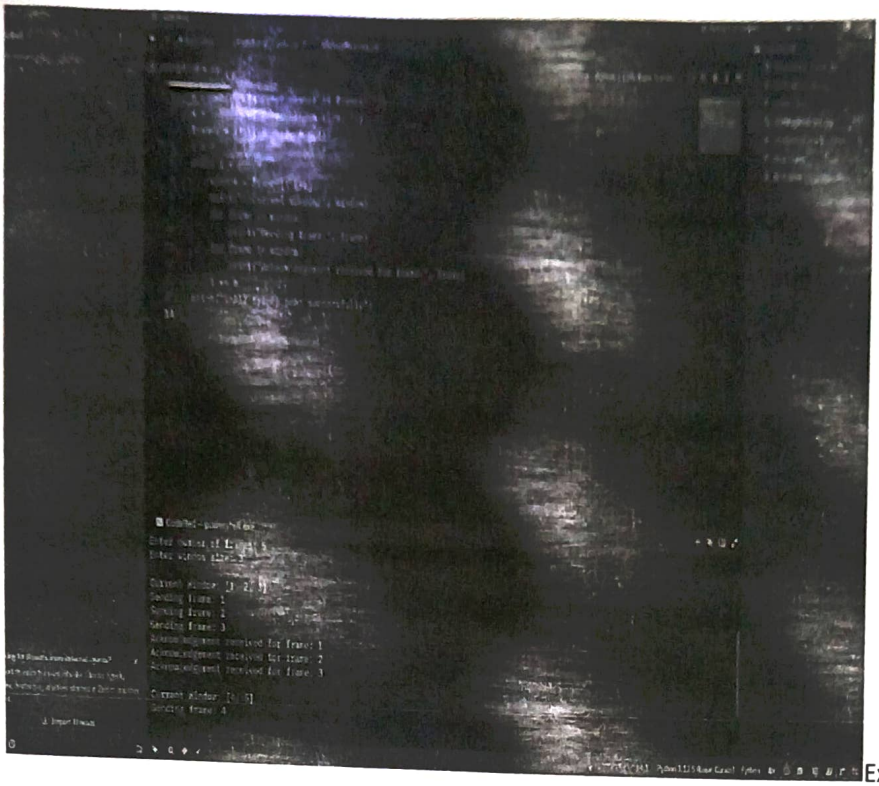
    base = 0
    nesct_frame = 0
    while base < total_frames:
        end = min (base + window_size - 1, total_fram
        Print (f"\n Sender Window : frames {base}
                                        -1)
                                        to {end}");
    Send = [ ]
    lost_now = false
    while nesct_frame <= end and nesct_frame <
                                    total frames :
        if nesct_frame == loss_frame :
            Print (f"Node A sends frame {nesct
                                        _frame
                                        -> lost)
            lost_now = True
        else :
            sent.append (nesct_frame)
        nesct_frame += 1
```

```
        ack = base-1
        for frame in sent:
            Print (f"Node B receive fra
                                    f fram
            ack = frame
            base += 1

        if ack >=0:
            Print (f'Node B sends ACK
        else: Print ("Node B sends no ACK")

        if last_now and base <= loss_fra
            Print ("Sender timeout -> Go-
            nesct_frame = base
            loss_frame = -1
        Print ("\n All frames delivered!")

if __name__ = "__main__":
        total = int (input ("Number of fra
        window = int (input ("Windowsize:"
        loss = int (input ())
        Sliding_window (total, windo
```



Experiment 7

Result :

The code Executed Successfully